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compact3

java.lang.management

Interface RuntimeMXBean

All Superinterfaces:

`PlatformManagedObject`

```
public interface RuntimeMXBean
extends PlatformManagedObject
```

The management interface for the runtime system of the Java virtual machine.

A Java virtual machine has a single instance of the implementation class of this interface. This instance implementing this interface is an MXBean that can be obtained by calling the `ManagementFactory.getRuntimeMXBean()` method or from the `platform MBeanServer` method.

The `ObjectName` for uniquely identifying the MXBean for the runtime system within an MBeanServer is:

```
java.lang:type=Runtime
```

It can be obtained by calling the `PlatformManagedObject.getObjectName()` method.

This interface defines several convenient methods for accessing system properties about the Java virtual machine.

Since:

1.5

See Also:

`ManagementFactory.getPlatformMXBeans(Class)`, `JMX Specification.`, `Ways to Access MXBeans`

Method Summary

Abstract Methods

Modifier and Type	Method	Description
`String`	`getBootClassPath()`	Returns the boot class path that is used by the bootstrap class loader to search for class files.
`String`	`getClassPath()`	Returns the Java class path that is used by the system class loader to search for class files.
`List<String>`	`getInputArguments()`	Returns the input arguments passed to the Java virtual machine which does not include the arguments to the `main` method.
`String`	`getLibraryPath()`	Returns the Java library path.
`String`	`getManagementSpecVersion()`	Returns the version of the specification for the management interface implemented by the running Java virtual machine.
`String`	`getName()`	Returns the name representing the running Java virtual machine.
`String`	`getSpecName()`	Returns the Java virtual machine specification name.
`String`	`getSpecVendor()`	

		Returns the Java virtual machine specification vendor.
String	getSpecVersion()	Returns the Java virtual machine specification version.
long	getStartTime()	Returns the start time of the Java virtual machine in milliseconds.
Map<String,String>	getSystemProperties()	Returns a map of names and values of all system properties.
long	getUptime()	Returns the uptime of the Java virtual machine in milliseconds.
String	getVmName()	Returns the Java virtual machine implementation name.
String	getVmVendor()	Returns the Java virtual machine implementation vendor.
String	getVmVersion()	Returns the Java virtual machine implementation version.
boolean	isBootClassPathSupported()	Tests if the Java virtual machine supports the boot class path mechanism used by the bootstrap class loader to search for class files.

Methods inherited from interface java.lang.management.PlatformManagedObject

getObjectName

Method Detail

getName

`String getName()`

Returns the name representing the running Java virtual machine. The returned name string can be any arbitrary string and a Java virtual machine implementation can choose to embed platform-specific useful information in the returned name string. Each running virtual machine could have a different name.

Returns:

the name representing the running Java virtual machine.

getVmName

`String getVmName()`

Returns the Java virtual machine implementation name. This method is equivalent to `System.getProperty("java.vm.name")`.

Returns:

the Java virtual machine implementation name.

Throws:

`SecurityException` - if a security manager exists and its checkPropertiesAccess method doesn't allow access to this system property.

See Also:

`SecurityManager.checkPropertyAccess(java.lang.String)`, `System.getProperty(java.lang.String)`

getVmVendor

`String getVmVendor()`

Returns the Java virtual machine implementation vendor. This method is equivalent to `System.getProperty("java.vm.vendor")`.

Returns:

the Java virtual machine implementation vendor.

Throws:

SecurityException - if a security manager exists and its checkPropertiesAccess method doesn't allow access to this system property.

See Also:

SecurityManager.checkPropertyAccess(java.lang.String), System.getProperty(java.lang.String)

getVmVersion

String getVmVersion()

Returns the Java virtual machine implementation version. This method is equivalent to System.getProperty("java.vm.version").

Returns:

the Java virtual machine implementation version.

Throws:

SecurityException - if a security manager exists and its checkPropertiesAccess method doesn't allow access to this system property.

See Also:

SecurityManager.checkPropertyAccess(java.lang.String), System.getProperty(java.lang.String)

getSpecName

String getSpecName()

Returns the Java virtual machine specification name. This method is equivalent to System.getProperty("java.vm.specification.name").

Returns:

the Java virtual machine specification name.

Throws:

SecurityException - if a security manager exists and its checkPropertiesAccess method doesn't allow access to this system property.

See Also:

SecurityManager.checkPropertyAccess(java.lang.String), System.getProperty(java.lang.String)

getSpecVendor

`String getSpecVendor()`

Returns the Java virtual machine specification vendor. This method is equivalent to `System.getProperty("java.vm.specification.vendor")`.

Returns:

the Java virtual machine specification vendor.

Throws:

`SecurityException` - if a security manager exists and its checkPropertiesAccess method doesn't allow access to this system property.

See Also:

`SecurityManager.checkPropertyAccess(java.lang.String)`, `System.getProperty(java.lang.String)`

getSpecVersion

`String getSpecVersion()`

Returns the Java virtual machine specification version. This method is equivalent to `System.getProperty("java.vm.specification.version")`.

Returns:

the Java virtual machine specification version.

Throws:

`SecurityException` - if a security manager exists and its checkPropertiesAccess method doesn't allow access to this system property.

See Also:

`SecurityManager.checkPropertyAccess(java.lang.String)`, `System.getProperty(java.lang.String)`

getManagementSpecVersion

`String getManagementSpecVersion()`

Returns the version of the specification for the management interface implemented by the running Java virtual machine.

Returns:

the version of the specification for the management interface implemented by the running Java virtual machine.

getClassPath

`String getClassPath()`

Returns the Java class path that is used by the system class loader to search for class files. This method is equivalent to `System.getProperty("java.class.path")`.

Multiple paths in the Java class path are separated by the path separator character of the platform of the Java virtual machine being monitored.

Returns:

the Java class path.

Throws:

`SecurityException` - if a security manager exists and its checkPropertiesAccess method doesn't allow access to this system property.

See Also:

`SecurityManager.checkPropertyAccess(java.lang.String)`, `System.getProperty(java.lang.String)`

getLibraryPath

`String getLibraryPath()`

Returns the Java library path. This method is equivalent to `System.getProperty("java.library.path")`.

Multiple paths in the Java library path are separated by the path separator character of the platform of the Java virtual machine being monitored.

Returns:

the Java library path.

Throws:

`SecurityException` - if a security manager exists and its checkPropertiesAccess method doesn't allow access to this system property.

See Also:

`SecurityManager.checkPropertyAccess(java.lang.String)`, `System.getProperty(java.lang.String)`

isBootClassPathSupported

```
boolean isBootClassPathSupported()
```

Tests if the Java virtual machine supports the boot class path mechanism used by the bootstrap class loader to search for class files.

Returns:

`true if the Java virtual machine supports the class path mechanism; false otherwise.`

getBootClassPath

```
String getBootClassPath()
```

Returns the boot class path that is used by the bootstrap class loader to search for class files.

Multiple paths in the boot class path are separated by the path separator character of the platform on which the Java virtual machine is running.

A Java virtual machine implementation may not support the boot class path mechanism for the bootstrap class loader to search for class files. The `isBootClassPathSupported()` method can be used to determine if the Java virtual machine supports this method.

Returns:

`the boot class path.`

Throws:

`UnsupportedOperationException - if the Java virtual machine does not support this operation.`

`SecurityException - if a security manager exists and the caller does not have ManagementPermission("monitor").`

getInputArguments

```
List<String> getInputArguments()
```

Returns the input arguments passed to the Java virtual machine which does not include the arguments to the `main` method. This method returns an empty list if there is no input argument to the Java virtual machine.

Some Java virtual machine implementations may take input arguments from multiple different sources: for examples, arguments passed from the application that launches the Java virtual machine such as the 'java' command, environment variables, configuration files, etc.

Typically, not all command-line options to the 'java' command are passed to the Java virtual machine. Thus, the returned input arguments may not include all command-line options.

MBeanServer access:
The mapped type of `List<String>` is `String[]`.

Returns:

`a list of String objects; each element is an argument passed to the Java virtual machine.`

Throws:

`SecurityException - if a security manager exists and the caller does not have ManagementPermission("monitor").`

getUptime

`long getUptime()`

Returns the uptime of the Java virtual machine in milliseconds.

Returns:

`uptime of the Java virtual machine in milliseconds.`

getStartTime

`long getStartTime()`

Returns the start time of the Java virtual machine in milliseconds. This method returns the approximate time when the Java virtual machine started.

Returns:

`start time of the Java virtual machine in milliseconds.`

getSystemProperties

`Map<String,String> getSystemProperties()`

Returns a map of names and values of all system properties. This method calls `System.getProperties()` to get all system properties. Properties whose name or value is not a `String` are omitted.

MBeanServer access:
The mapped type of `Map<String,String>` is `TabularData` with two items in each row as follows:

Item Name	Item Type

key String

value String

Returns:

a map of names and values of all system properties.

Throws:

SecurityException - if a security manager exists and its checkPropertiesAccess method doesn't allow access to the system properties.

Submit a bug or feature

For further API reference and developer documentation, see Java SE Documentation. That documentation contains more detailed, developer-targeted descriptions, with conceptual overviews, definitions of terms, workarounds, and working code examples.

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